Eigenvalues and eigenvectors: the directions a matrix only stretches

In a table with many columns, which directions do the data vary along most, and how do we look at embeddings with hundreds of numbers?

Mathematics for AI and Machine Learning | Session 5 of 18 | Linear algebra

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Today in one view: what you will leave with

By the end of today you can



Say what an eigenvector and an eigenvalue are, with a picture of a matrix acting on a circle

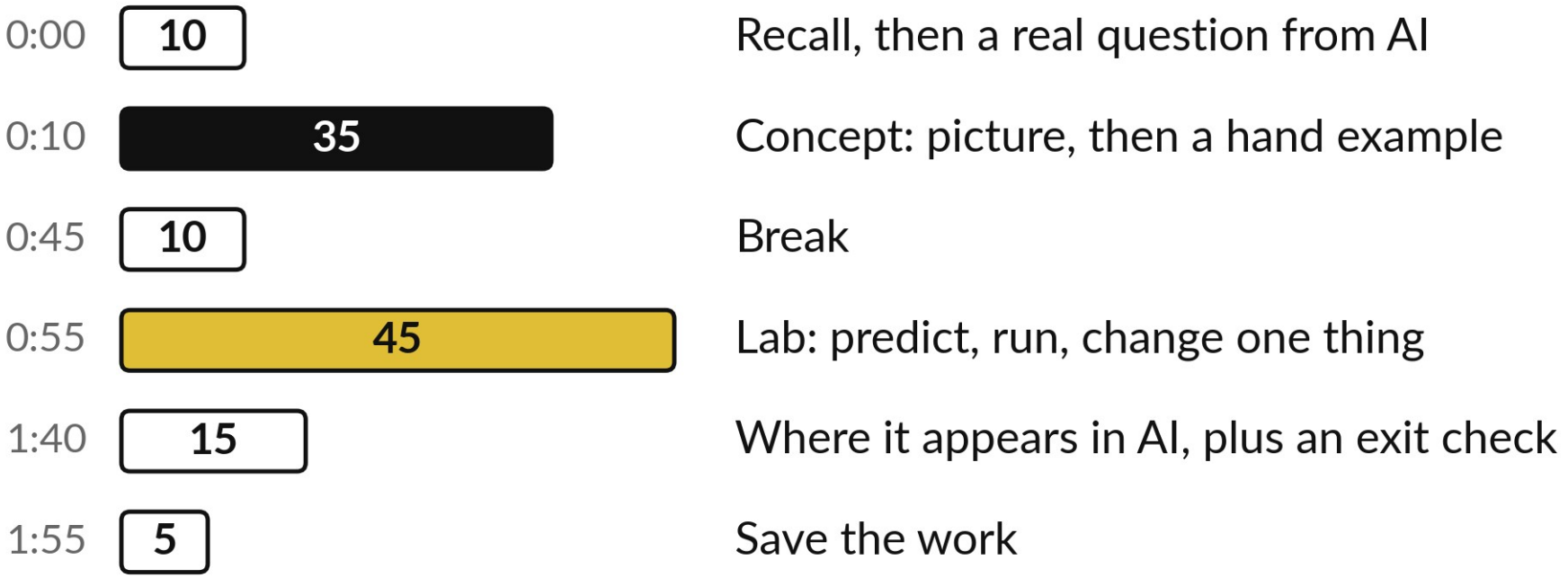


Find the eigenvalues and eigenvectors of a 2 by 2 matrix by hand and check them with trace and determinant



Build a covariance matrix, find its top direction, and use power iteration to find it by repeated multiplying

Your two hours: stretch directions, then data



Ideas first, with pictures. Then the lab in Python or in a spreadsheet. Then where it shows up in AI.

Three questions from last time

1. What does a determinant of 0 tell you about a matrix?

2. Write the normal equation for fitting a line.

3. What is the rank of the matrix with rows (1, 2) and (2, 4)?

Answer aloud first. Then we check together.

Answers to the three questions from Session 4

Question	Short answer
What does a determinant of 0 tell you about a matrix?	It squashes the plane onto a line or a point. It has no inverse and its rank is below full.
Write the normal equation for fitting a line.	$X^TX\,w = X^Ty$. It gives the weights w with the smallest squared error.
What is the rank of the matrix with rows (1, 2) and (2, 4)?	1. The second column is twice the first, so the columns give only one direction.

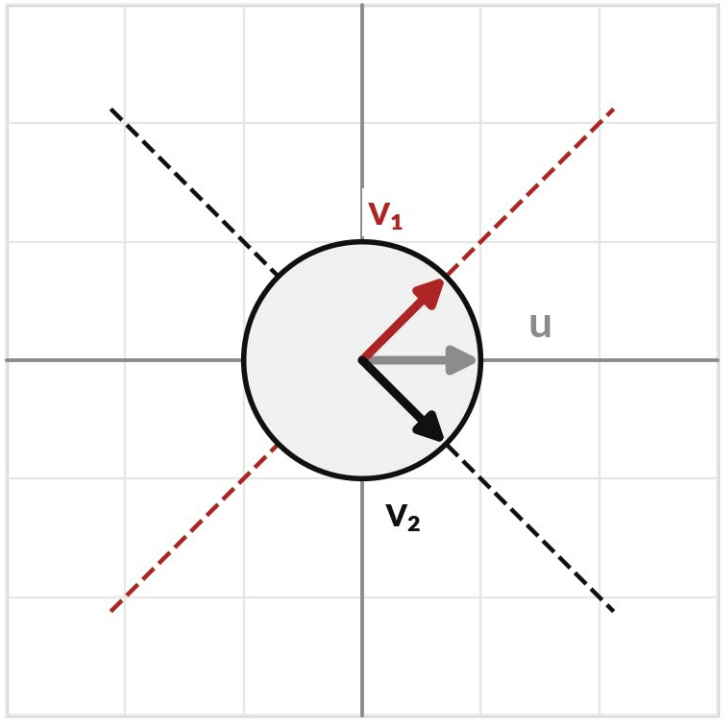
If one surprised you, look back at Session 4 before the lab. Rank and determinant return today.

Words we use today

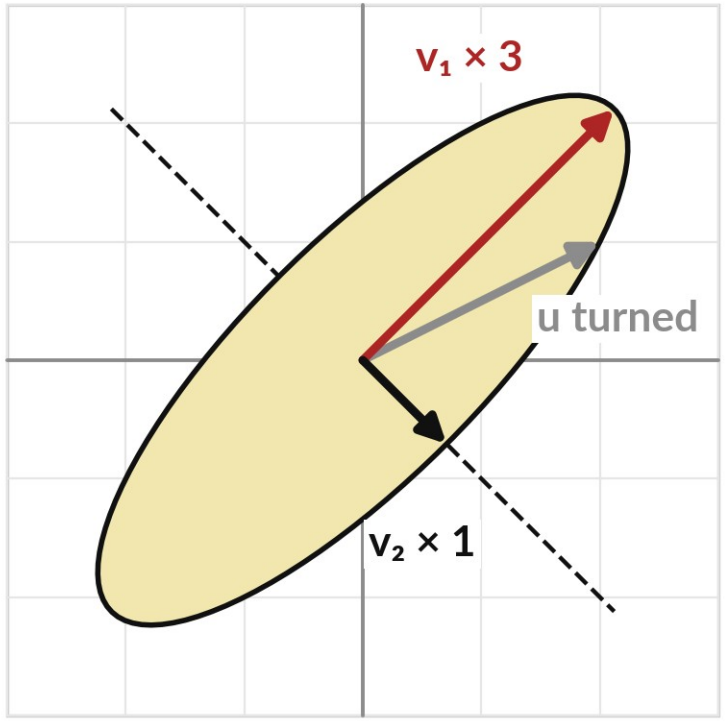
	Word	What it means
	**Eigenvector**	A direction that a matrix only stretches, flips or squashes. It stays on its own line.
	**Eigenvalue**	The factor by which an eigenvector is stretched. Written with the Greek letter lambda (λ).
	**Trace**	The sum of the diagonal entries of a square matrix.
	**Covariance matrix**	A table of how every pair of features varies together. The diagonal holds each feature's variance, the typical squared distance from its mean.
	**Power iteration**	Multiplying by a matrix again and again to find its top eigenvector.

Most arrows turn when a matrix acts on them, but a few directions only stretch

Before: the unit circle



After the matrix: an ellipse



The matrix with rows (2, 1) and (1, 2) turns a circle into an ellipse. The grey arrow u turns.

v_1 points along (1, 1) and becomes 3 times longer.

v_2 points along (1, −1) and keeps its length.

Such directions are eigenvectors.

An eigenvector v and its eigenvalue λ satisfy Av = λv

Definition

$$A\mathbf{v} = \lambda\mathbf{v}$$

Our first pair

$$A\,(1, 1) = (3, 3) = 3\,(1, 1)$$

Symbol	What it means	In our data
A	A square matrix. Here the rows are (2, 1) and (1, 2).	2 by 2
v	An eigenvector: a nonzero vector whose direction A keeps.	(1, 1) or (1, −1)
λ	The eigenvalue (Greek letter lambda): the factor by which A stretches v.	3 for (1, 1), and 1 for (1, −1)
Av	Matrix times vector. For (1, 1): (2 + 1, 1 + 2).	(3, 3)

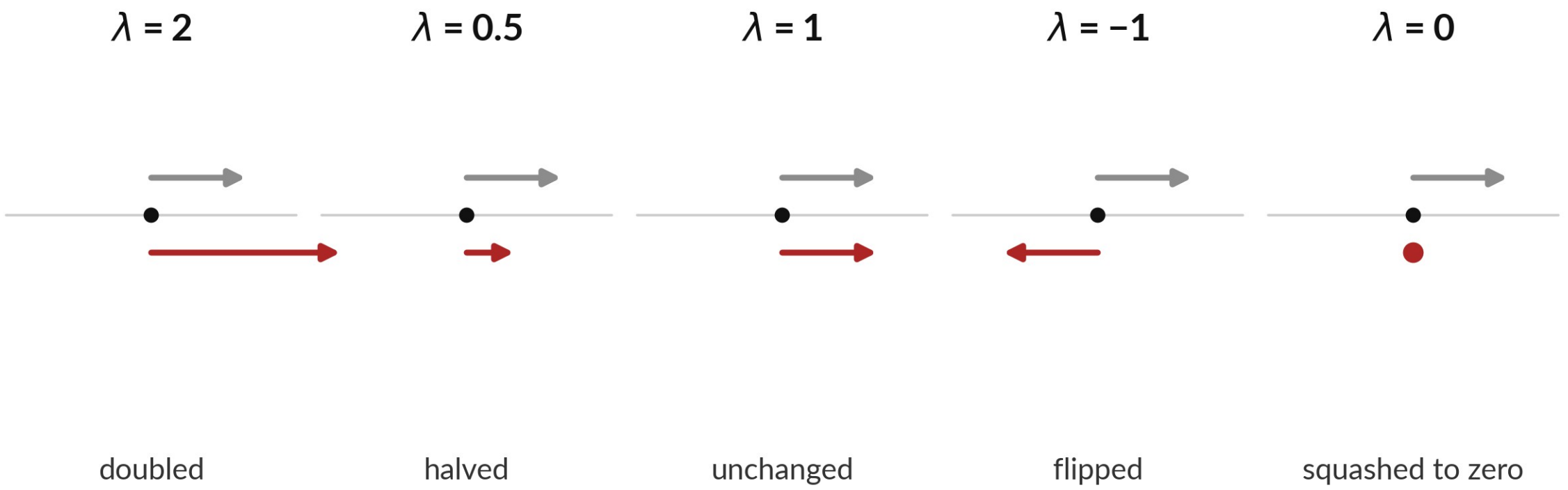
A negative λ flips v. A λ of 0 squashes it to the origin. Any multiple of v is an eigenvector too.

Testing four arrows by hand: only two directions survive

Arrow v	Av	Same line as v?	Stretch factor
(1, 0)	(2, 1)	No, it turned	none
(1, 1)	(3, 3)	Yes	3
(1, -1)	(1, −1)	Yes	1
(1, 2)	(4, 5)	No, it turned	none

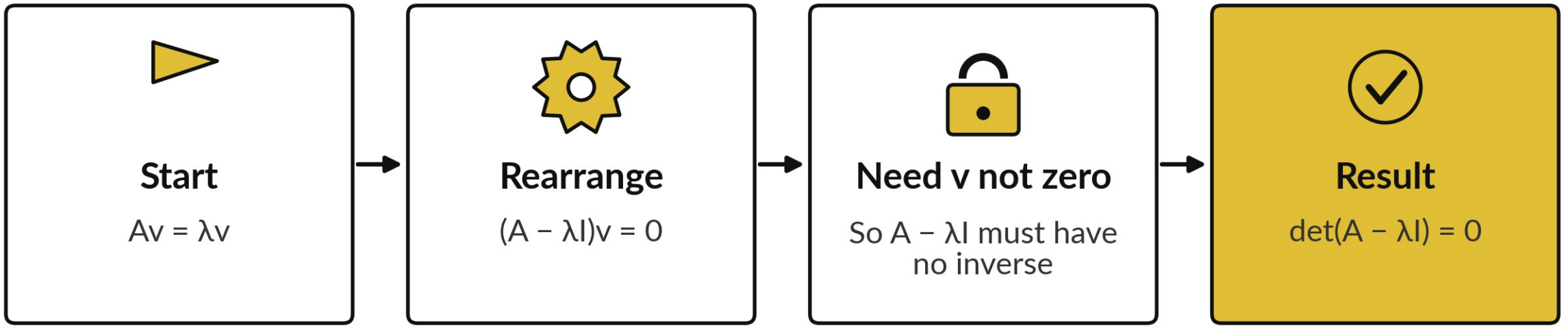
A has rows (2, 1) and (1, 2). Compute Av with a dot product for each row: for (1, 2) it is 2 + 2 = 4 and 1 + 4 = 5.

The eigenvalue says whether the arrow is stretched, shrunk, flipped or squashed



Grey arrow: an eigenvector. Red arrow: what A turns it into. λ (lambda) is the multiplier.

To find eigenvalues, ask when A - λI has no inverse



λI has λ on the diagonal. If A − λI had an inverse, multiplying (A − λI)v = 0 by it would give v = 0. So no inverse: determinant 0 (Session 4).

Writing det(A - λI) = 0 out for our matrix

A has rows (2, 1) and (1, 2). We subtract λ from each diagonal entry and take the determinant.

A - λI	Rows $(2 - \lambda, 1)$ and $(1, 2 - \lambda)$.
Determinant	$(2 - \lambda)(2 - \lambda) - 1 \times 1 = 4 - 4\lambda + \lambda^2 - 1 = \lambda^2 - 4\lambda + 3$.
Set it to 0	$\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0$, so $\lambda = 3$ or $\lambda = 1$.
Any 2 by 2	The same steps give $\lambda^2 - T\lambda + D = 0$, with T the sum of the diagonal and D the determinant.

For any 2 by 2 matrix the characteristic equation is a quadratic

Characteristic equation

$$\lambda^2 - T\lambda + D = 0$$

Eigenvalues

$$\lambda = \frac{T \pm \sqrt{T^2 - 4D}}{2}$$

Symbol	What it means	In our data
T	The trace: the sum of the diagonal entries, $a_{11} + a_{22}$.	2 + 2 = 4
D	The determinant: $a_{11}a_{22} - a_{12}a_{21}$ (a_{ij} is row i, column j of A).	2 × 2 − 1 × 1 = 3
±	Plus gives one eigenvalue, minus gives the other.	(4 + 2) / 2 and (4 − 2) / 2
λ	The two eigenvalues.	3 and 1

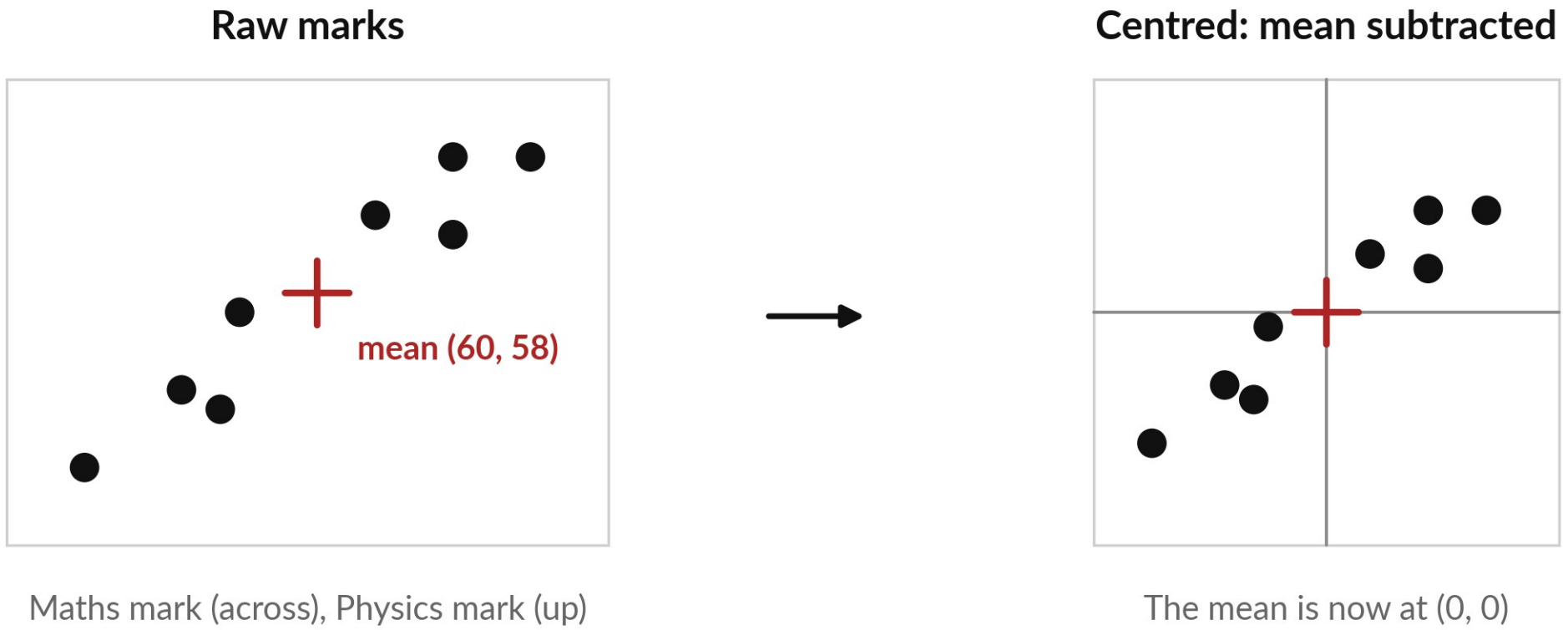
Check: the eigenvalues add up to the trace (3 + 1 = 4) and multiply to the determinant (3 × 1 = 3).

Worked example: eigenvalues and eigenvectors of our matrix

A has rows (2, 1) and (1, 2). We find λ first, then an eigenvector (s, t) for each λ.

Eigenvalues	$\lambda^2 - 4\lambda + 3 = 0$. So $\lambda = (4 + 2) / 2 = 3$ or $\lambda = (4 - 2) / 2 = 1$.
For λ = 3	$A - 3I$ has rows $(-1, 1)$ and $(1, -1)$. The equation $-s + t = 0$ gives $s = t$, so $v_1 = (1, 1)$.
For λ = 1	$A - I$ has rows $(1, 1)$ and $(1, 1)$. The equation $s + t = 0$ gives $v_2 = (1, -1)$.
Check	$A(1, 1) = (3, 3)$ and $A(1, -1) = (1, -1)$. Both match.

Centring: subtract each feature's mean so the cloud sits on the origin



Eight students (illustrative). Mean maths 60, mean physics 58. Each mark minus its mean.

Mean, variance and covariance measure the centre, the spread and the tilt

Variance (sample)

$$\text{var}(x) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Covariance (sample)

$$\text{cov}(x, y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Symbol	What it means	In our data
n, x_i, y_i	n students; x_i and y_i are the marks of student i. Σ adds them up.	8 students
$\bar{x}$ (x-bar)	The mean: (x_1 + ... + x_n) / n. Later sessions also write it μ.	60 and 58
var, cov	Variance: typical squared distance from the mean. Covariance: positive when both rise together.	66, 38; 48
n - 1	Distances to a mean taken from the same data come out a little small; n − 1 corrects that.	divide by 7

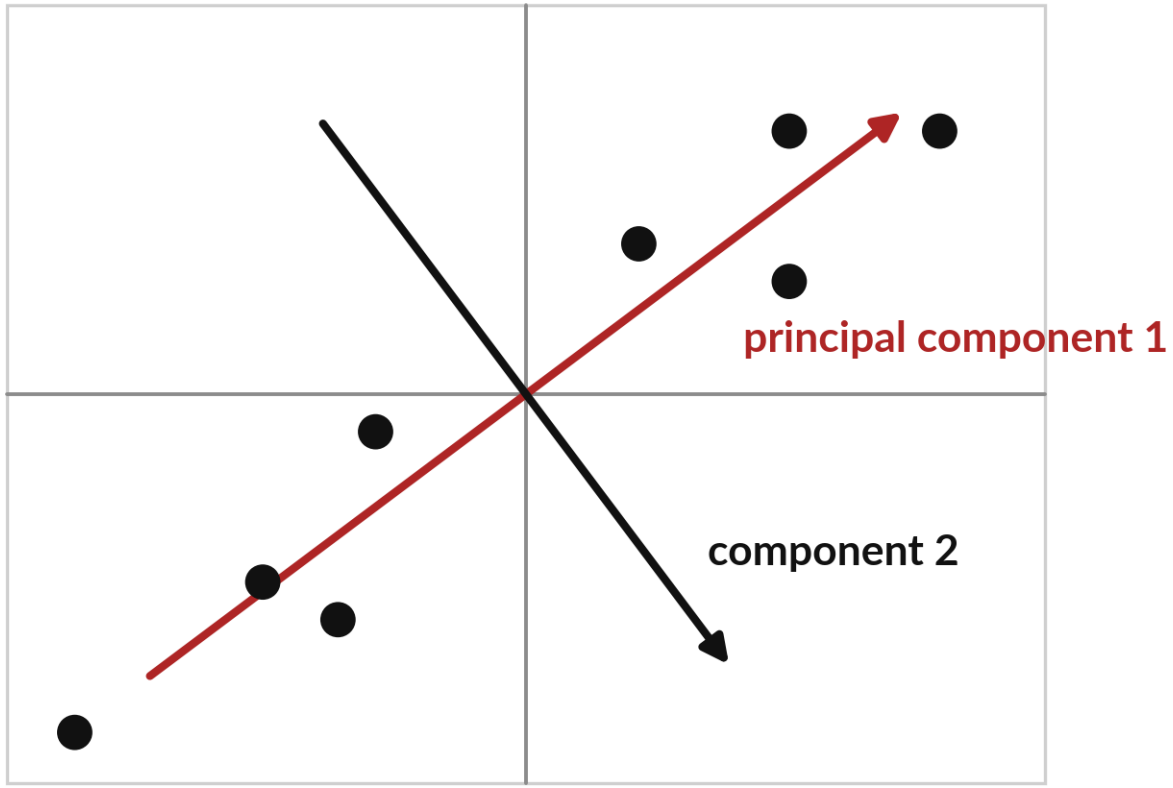
Sample version: np.cov, VAR.S, COVARIANCE.S. Dividing by n gives 57.75, 33.25, 42. Session 13 returns to this.

Worked by hand: the covariance matrix of the eight marks

Illustrative marks. Centred = mark minus its mean. In matrix form $C = Xc^{T} Xc / (n - 1)$, with Xc the centred marks.

Maths marks	53, 55, 63, 67, 48, 67, 56, 71. Mean 60. Centred: −7, −5, 3, 7, −12, 7, −4, 11.
Physics marks	53, 52, 62, 65, 49, 61, 57, 65. Mean 58. Centred: −5, −6, 4, 7, −9, 3, −1, 7.
Sums	Squared centred maths marks add to 462, physics to 266. Maths times physics: 336.
Divide	Divide by n − 1 = 7. Variances 66 and 38. Covariance 48. So C has rows (66, 48) and (48, 38).

The eigenvectors of the covariance matrix are the directions of the cloud



Centred maths mark (across) and centred physics mark (up)

Centred students. Red: the top eigenvector v_1 of C. Black: v_2.

v_1 = (0.8, 0.6), variance 102: the direction of greatest spread, called the first principal component (Session 6).

v_2 = (0.6, −0.8): variance 2.

Each eigenvalue is the variance along its eigenvector.

Worked by hand: eigenvalues and top eigenvector of C

C has rows (66, 48) and (48, 38). We use the 2 by 2 formulas from earlier.

Trace and determinant	T = 66 + 38 = 104. D = 66 × 38 − 48 × 48 = 204.
Eigenvalues	The square root of $104^2 - 4 \times 204$ is 100. So λ = (104 + 100) / 2 = 102 or (104 − 100) / 2 = 2.
Top eigenvector	C − 102I has rows (−36, 48) and (48, −64). So −36s + 48t = 0, which gives t = 0.75s. Take (4, 3). Length 5, so v_1 = (0.8, 0.6).
Share of variance	102 / 104 = 98.1 percent of all the variance lies along direction 1.

Power iteration finds the top eigenvector by repeated multiplying

Update

$$\mathbf{g}_{k+1} = \frac{A\,\mathbf{g}_k}{\max|A\,\mathbf{g}_k|}$$

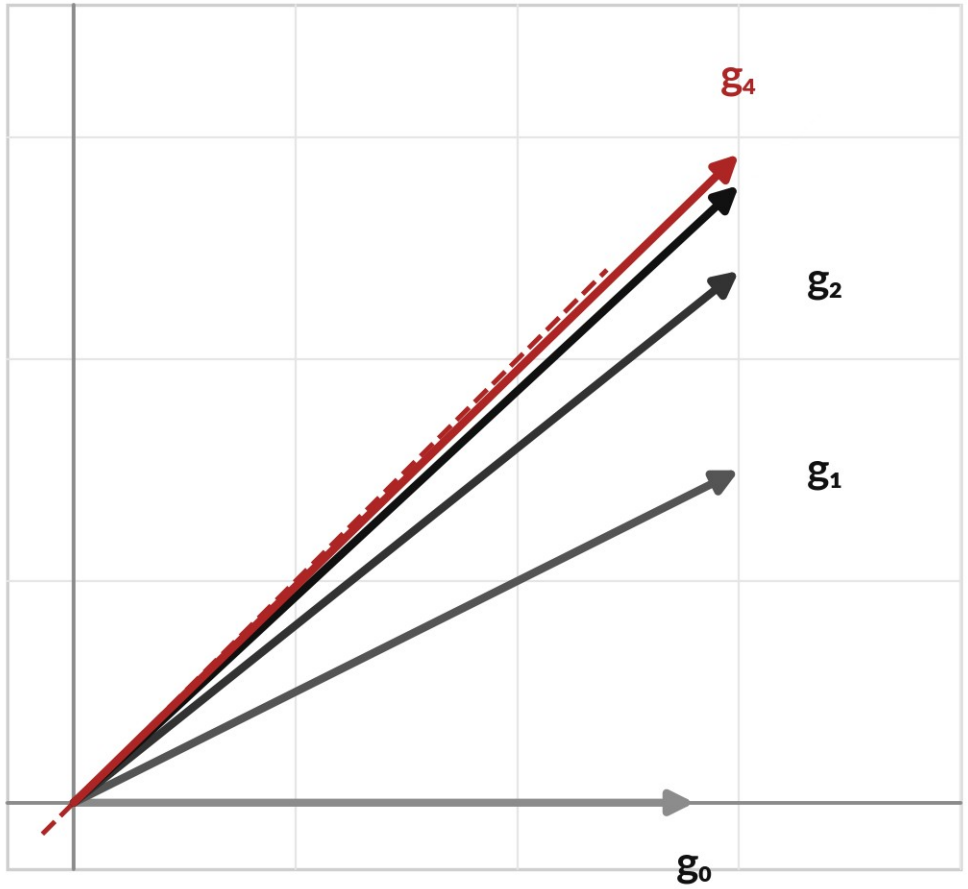
Eigenvalue estimate

$$\lambda \approx \max|A\,\mathbf{g}_k|$$

Symbol	What it means	In our data
$\mathbf{g}_k$	The guess after k steps. Almost any start works.	g_0 = (1, 0) here
$\mathbf{A}\mathbf{g}_k$	Multiply the guess by the matrix.	one matrix times vector product
max |.|	The entry with the largest size. Dividing by it keeps numbers small.	g_k has largest entry 1
λ	Top eigenvalue: $\max\|Ag_k\|$ estimates it because g_k has largest entry 1.	estimate tends to 3

It works when one eigenvalue is largest in size. When the top two are nearly equal in size, it converges slowly.

Each multiplication swings the arrow closer to the top eigen-direction



Matrix with rows (2, 1) and (1, 2). Start at g_0 = (1, 0). Multiply, then divide by the largest entry.

g_1 = (1, 0.50), g_2 = (1, 0.80), g_4 is almost (1, 1).

The estimates of λ are 2.00, 2.50, 2.80, 2.93, 2.98. They head to 3.

Power iteration by hand on the covariance matrix C

Step k	Guess g_k	C g_k	Largest entry
0	(0.000, 1.000)	(48.0, 38.0)	48.00
1	(1.000, 0.792)	(104.0, 78.1)	104.00
2	(1.000, 0.751)	(102.0, 76.5)	102.04
3	(1.000, 0.750)	(102.0, 76.5)	102.00
4	(1.000, 0.750)	(102.0, 76.5)	102.00
5	(1.000, 0.750)	(102.0, 76.5)	102.00

This time the start is g_0 = (0, 1), to show that any start works. After 6 multiplications (row k = 5) the estimate is 102.0, close to 102, and the guess is near (1, 0.75), which is (0.8, 0.6) at length 1.

Four things people get wrong about eigenvectors

Mistake	What is true	What to do
Treating an eigenvector as one fixed arrow	Any multiple, and the opposite arrow, is also an eigenvector.	Quote a unit-length version. Expect sign flips between tools.
Looking for eigenvalues of a non-square matrix	Only square matrices have them.	Use the SVD for other shapes (Session 6).
Expecting real eigenvalues always	A quarter-turn rotation has no real eigenvalues, because every arrow turns.	Covariance matrices are symmetric and always give real ones.
Mixing up the two outputs	For a covariance matrix, the eigenvalue is the spread along its eigenvector.	Quote both together.

Centre, divide by n - 1, then eigh: the covariance route in five lines

```
Xc = X - X.mean(axis=0)
C = Xc.T @ Xc / (len(X) - 1)
values, vectors = np.linalg.eigh(C)
g = np.array([0.0, 1.0])
for k in range(6):
    Cg = C @ g
    est = np.max(np.abs(Cg))
    g = Cg / est
```

In plain words

Xc = ... centres each column by subtracting its mean.

C = ... is the covariance matrix, dividing by n − 1 (the same as np.cov(X.T)).

eigh is for symmetric matrices. It lists eigenvalues from smallest to largest, with eigenvectors in the columns.

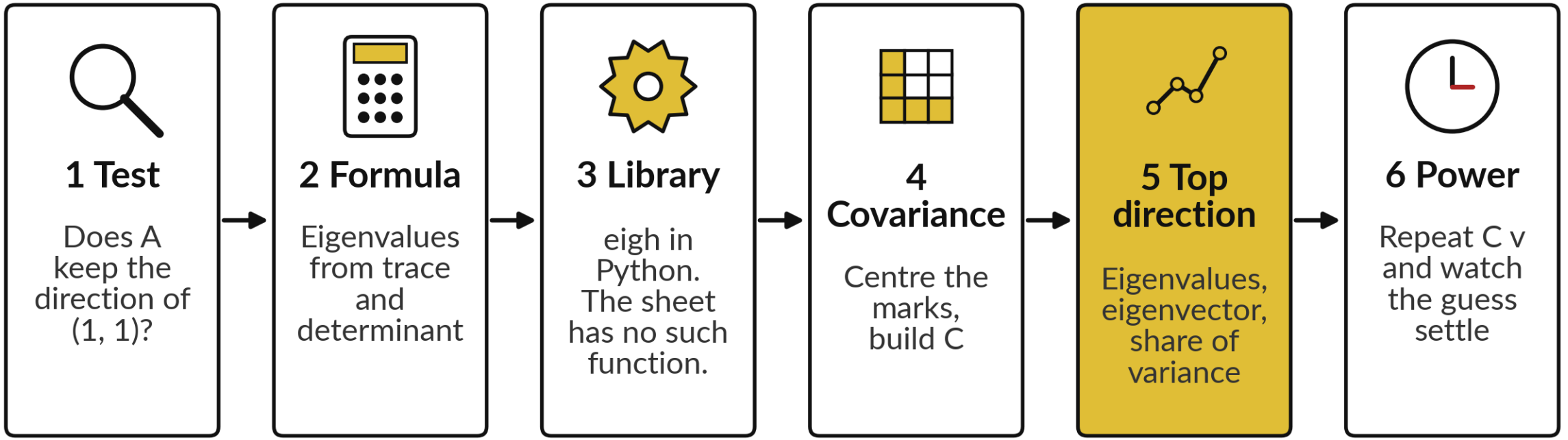
The loop from g = ... is power iteration. est holds the eigenvalue estimate.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

Six lab steps: one arrow, two formulas, then the marks of eight students



Python uses numpy. The spreadsheet uses MMULT, VAR.S, COVARIANCE.S, SQRT and rows of cells. Gold cells are yours.

Check your lab against 3 and 1, then C, then 102 and 2

What you compute	You should see
A times (1, 1) for the small matrix	(3, 3)
Eigenvalues of the small matrix	3 and 1
Covariance matrix C of the eight marks (divide by n − 1)	rows (66, 48) and (48, 38)
Eigenvalues of C	102 and 2
Top eigenvector of C, unit length	(0.8, 0.6)
Share of variance along direction 1	0.981
Power iteration after 6 multiplications (row k = 5): largest entry	102.0
Covariance if physics is replaced by 100 minus physics	−48: the sign flips, the size stays

If a number differs, stop and find the step that went wrong. The sign of an eigenvector may differ between tools.

Minus signs, ordering and centring explain most lab surprises

Error or symptom	What it says	What to do
The eigenvector has a minus sign	The opposite arrow is also an eigenvector.	Compare absolute values, or flip the sign.
Eigenvalues come out in the wrong order	eigh lists smallest first.	Reverse the list or sort it.
Covariance far too large	The data were not centred first.	Subtract the column means.
#NUM! in the spreadsheet	The square root of a negative number.	Check $T^2 - 4D$. For covariance it is never negative.

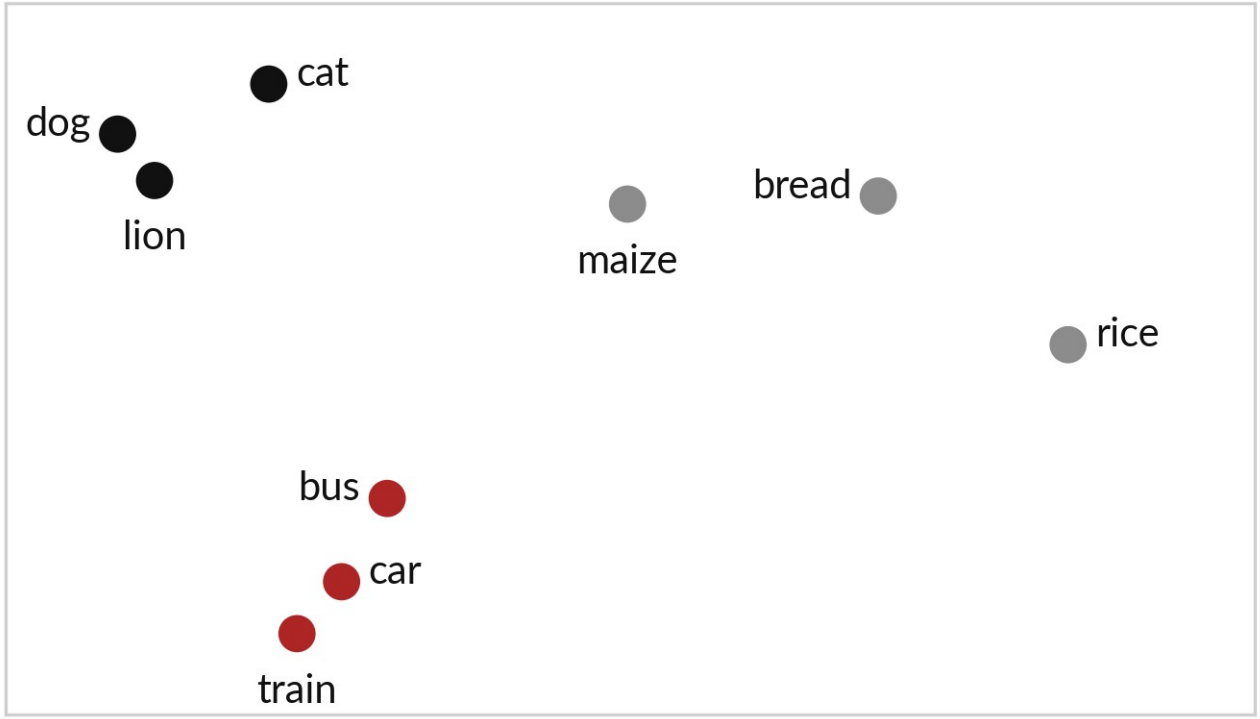
Where this appears in AI: PCA turns embeddings into a map you can look at

Each word is a list of numbers

(illustrative: 4 numbers, made up)

cat	1.00	1.05	0.20	−0.19
car	−0.12	0.15	0.76	0.88
rice	−0.31	−0.87	0.15	−0.93

PCA keeps the two directions with the most spread



principal component 1 (across) and 2 (up)

PCA (principal component analysis) of made-up 4-number vectors. Real embeddings are far longer.

Where this appears in AI: looking at what models learn



PCA keeps the top eigenvectors of the covariance matrix of the data.



The Embedding Projector, published by Google researchers in 2016, offers PCA to view embeddings.



Open-weight models such as Llama publish their weights, so this analysis can be run on them.



Closed assistants do not publish their internal embedding tables. The vectors a public embedding service returns for your own text can be projected the same way.

Where this appears in AI: ask an assistant for eigenvalues you already know

8 minutes

Hands-on, with any AI assistant

Ask an assistant for eigenvalues, then check them.

Ask for the eigenvalues and eigenvectors of the matrix with rows (2, 1) and (1, 2). Expect 3 and 1.

Ask again for the covariance matrix with rows (66, 48) and (48, 38). Expect 102 and 2.

Ask it to show the characteristic equation. Never paste confidential data.

You keep

A note of which answers matched, and any sign differences.

Exit check: Av = λv, two eigenvalues, and the top direction of data

Question	Short answer
What does Av = λv say in words?	A only stretches, flips or squashes the vector v. λ is the factor.
What are the eigenvalues of the matrix with rows (2, 1) and (1, 2)?	T = 4 and D = 3, so λ = 3 and 1.
What does the top eigenvector of a covariance matrix tell you?	The direction in which the data vary most. Its eigenvalue is the variance there.

Write your own answer first, then compare. Your lab checks are part of the exit check.

Save the lab and keep the four eigenvalues

What you keep from today

Keep the saved lab and four numbers: eigenvalues 3 and 1 for the small matrix, and 102 and 2 for the covariance matrix of the eight marks.

Our question today

In a table with many columns, which directions do the data vary along most, and how do we look at embeddings with hundreds of numbers?

Next session

Session 6: singular value decomposition and PCA, which build on today.

Take away

Saved lab
Notebook or workbook, saved

Small matrix
Eigenvalues 3 and 1

Covariance matrix
Eigenvalues 102 and 2

Next session
SVD and PCA